enviroshore

Oil solution specialists


Enviroshore. Custom-made oil solutions for our planet.

Sludge and Pond Lagoon Remediation

With an holistic approach to environmentally friendly removal and recovery of waste oil, Enviroshore is fully equipped to remove, transport and safely recycle slop/sludge waste oil through a specialised treatment facility into a usable product, thus reducing carbon emissions. This win-win process effectively alleviates a range of problems while frequently offering financial gains to the customer.

All stakeholders involved are issued a certificate of safe disposal for all waste slop/sludge oil discharged by the vessel.

Incorporating our specialized technologies, solutions and products we have developed a full cradle-to-grave solution for sludge ponds/lagoons whereby hydrocarbons are recovered and soil and ground water is remediated with minimal waste needing to go to costly hazardous landfills.

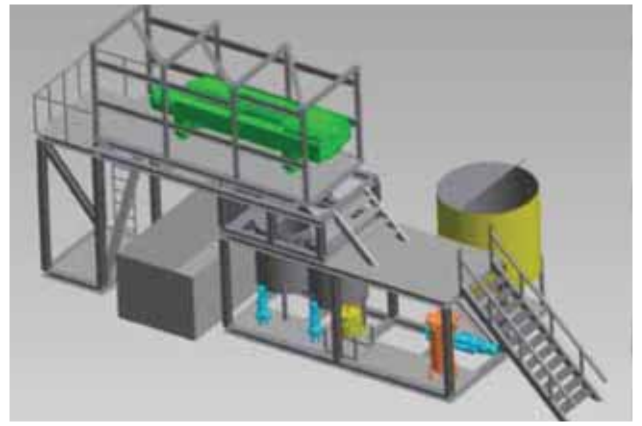
The Enviroshore services are available at most of the major Ports in both South Africa and Mozambique.

The SLOPO® system from Oreco

The SLOPO® system from Oreco offers efficient and innovative slop oil treatment to convert a wide range of slop oils into valuable products. Slop oil can cause dangerous environmental hazards and costly storage problems and poses a serious burden to oil companies, governments and communities alike. With SLOPO®, those issues are not only solved, they are transformed into profit.

That's where SLOPO® comes in

The SLOPO® system is designed to treat slop oils with a wide range of characteristics, from slop oil with a high solid content to slop oil from the desalter rag layer. The core technology of the SLOPO® system is centrifugal separators, used to separate components with different specific gravity. This special technology enables efficient, cost-effective slop oil recovery and the creation of marketable products. The unit is manufactured in standard sized containers, and is customised for use on remote sites in harsh environments. SLOPO® is made in a standard configuration which then can be applied to any capacity necessary.

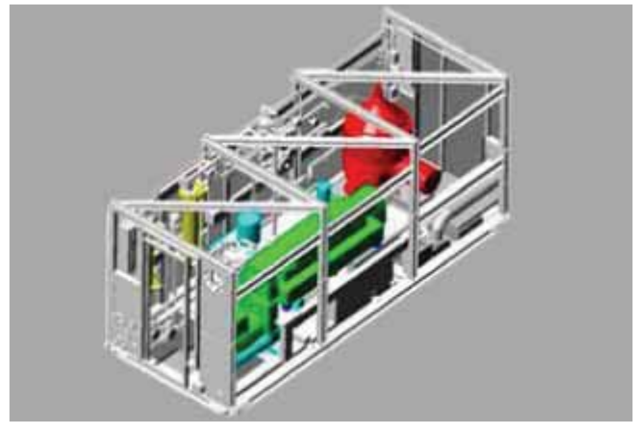


SLOPO® Type 1

The centrifugal force

SLOPO® is built in very compact modules in order to facilitate easy trans-portation to the site. The SLOPO® system is essentially plug and play with an integrated control system, insulated containers and heat traced piping. SLOPO® is constructed for use in hazardous areas, fully HAZOP'ed.

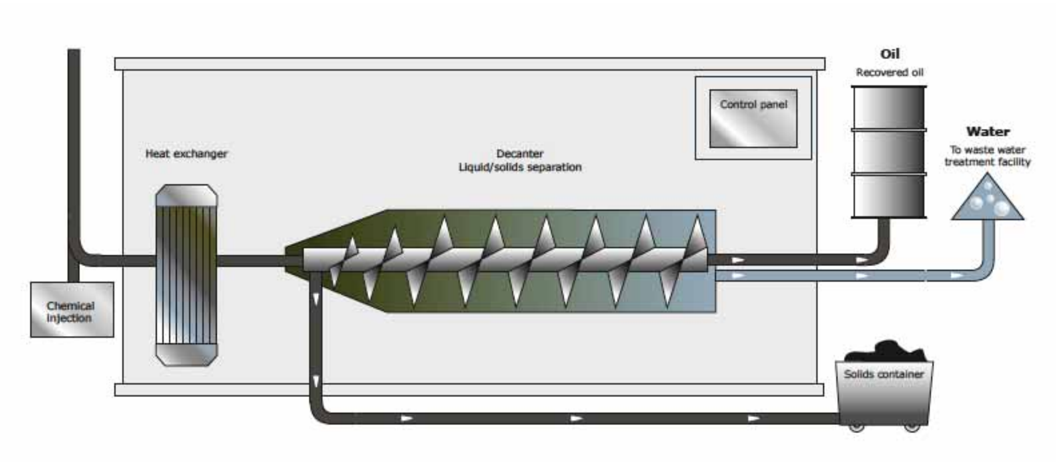
The core technology encompassed in the SLOPO® system is centrifu-gal separators, which have been applied in the oil/gas business for decades. The centrifugal force is used to separate components with different specific gravity.



SLOPO® Type 2

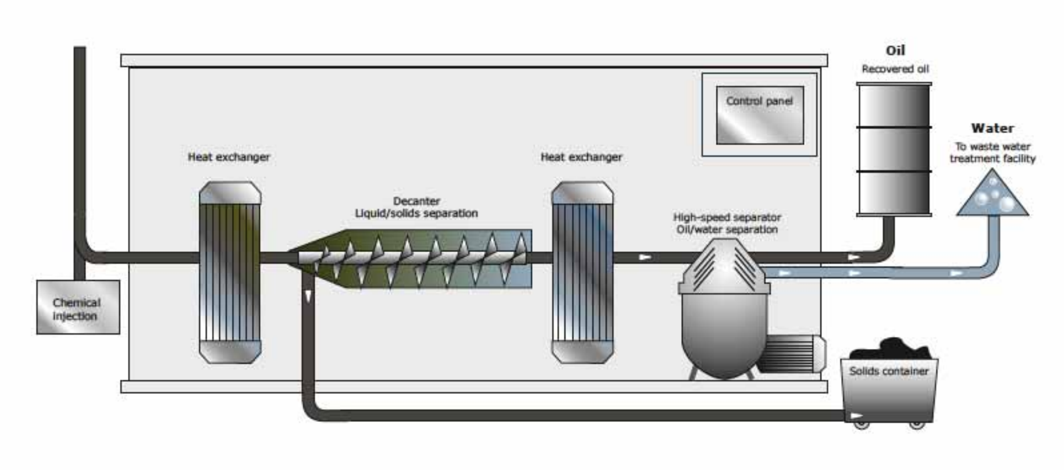
Slop Oil Systems

The SLOPO® system is made in two configurations, depending on the demand of the final quality of the effluent streams:



SLOPO® Type 1

This setup features a 3-phase decanter. This is combined with a very robust control system and an online interface adjustment which optimises the function of the system.



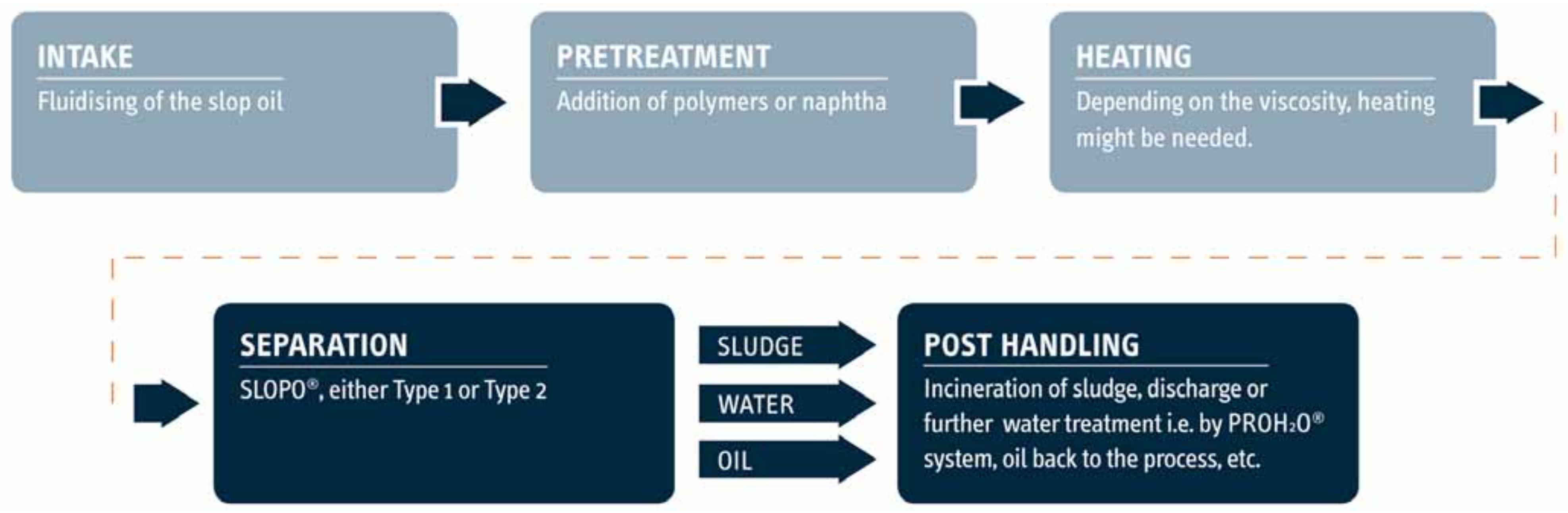
SLOPO® Type 2

This is a combination of two centrifuge technologies, a two-phase decanter and a high speed separator. This combination will be able to handle big variations in the slop oil with unchanged high quality of the effluents

Technical Specifications

	SLOPO® Type 1	SLOPO® Type 1
Technology	3-phase decanter	2-phase decanter and a high speed separator
Feed flow	10 m3/h	10 m3/h
Effluent water*	<1% oil	< 50 ppm oil
Effluent oil*	<1% BS&W	<1% BS&W
Effluent solids*	~50% TSS, < 1% oil	~50% TSS, < 1% oil
Physical properties	Decanter fits in a 20' container, whole installation can be in one 20' and one 40' container. Remote control in safe area Simple control	Compact, fits in a 20' container. Controls are integrated in the system. Many control possibilities.

Oil Flow Diagram



Pretreatment Possibilities

OIL PARAMETER	Explanation	System add-on
For slop oil with small particles of TSS	Chemical injection of polymers will flocculate the particles, thus making them more susceptible for the centrifugal forces and separate easier.	Oreco can deliver a dosing system which can handle both flocculants and demulsifiers.
For emulsified oil	Demulsifiers can be added in order to concentrate the water into larger droplets, thus increasing the gravity difference and thereby assisting the separation.	Oreco can deliver a dosing system which can handle both flocculants and demulsifiers.
For oils with SG larger than 0.95	When the different components in the slop oil have similar specific gravity, the centrifuges will not be able to separate the components.	If necessary, naphta can be added to the feed stream to dilute the oil, and thereby make it lighter. As water and solids keep their SG, the centrifuges will work.

Waste Oil Processing Facilities

Oily residues in the form of sludge from oil refineries and intermediate storage ('oil seas', "sludge lagoons') as well as oil-water mixtures from ships, discarded lubricant oils and emulsions or other waste water streams containing oil need not be considered troubling waste. They can now be utilised as a source of energy through oil recovery.

Enviroshore has established state-of-the-art mobile waste oil processing plants. These plants not only offer a solution that provides optimum efficiency in oil recovery, but also include waste disposal, exhaust air purification and waste water treatment. Enviroshore has successfully incorporated the organic waste oil treatment plant into its core business of developing unique environmental solutions.

The plants are situated in the Port of Motola in Mozambique and Cato Ridge in KwaZulu-Natal and are equipped to produce up to 10 000 tons of finished product per month. Enviroshore Global Mozambique Limited operates under strict quality control systems so the waste oil processing plant produces both high grade, quality heavy fuel oil and light fuel oil products which meet international standards. It is an energy efficient plant that maximizes all forms of energy produced and by incorporating our environmental technologies and products there is minimal, if any, waste for disposal. All waste water is treated using our enzyme and microbial technology, where all solids go through a remediation process and all vapour is recovered using vapor recovery technology. The waste oil processing plants are designed for the added advantage of custom-tailoring for each and every waste oil recovery project.

Waste oil processing is a highly efficient and environmentally-friendly oil recovery process which is at the core of our vision "To become the Principal partner of choice in Africa in all areas of rehabilitation of on-site storage clean up and soil remediation incorporating our specialised waste oil processing facility."

Business Concept

- A self contained "Recovery Centre" that recovers the waste hydrocarbon material.
  - Hydrocarbon Oils are brought to the storage facility by Vacuum Tankers/Road Tankers.
  - Hydrocarbon Oils are stored in a Bulk Tanker.
  - Product is efficiently controlled and monitored on a daily basis
  - Waste Hydrocarbon material is recovered offsite at our Oil Processing Facility.
  - Water is treated by means of our specialised environmental products
  - All On Site Storage Facilities will be operated by Enviroshore representatives, allowing refinery to focus on their core business function

Oil Recovery

Project Vision

Improved, efficient and environmentally-friendly Hydrocarbon recovery at any Refinery

Goal and Objective

- Allow the refinery to focus on their core business function, refining, whilst incorporating our specialised technology for oil recovery
- Make the latest available Oil recovery technology available to partners
- To develop an Efficient and Environmentally Friendly Facility.
- To Add Value and Services to the Refinery
- To Reduce Costly Waste Disposal of Hydrocarbon materials
- To upgrade the Environmental Status of the refinery
- Creation of a common vision between the refinery & Stakeholders
- Create sustainability through sound environmental and social practice
- Add significant value to the refinery, both economically and environmentally

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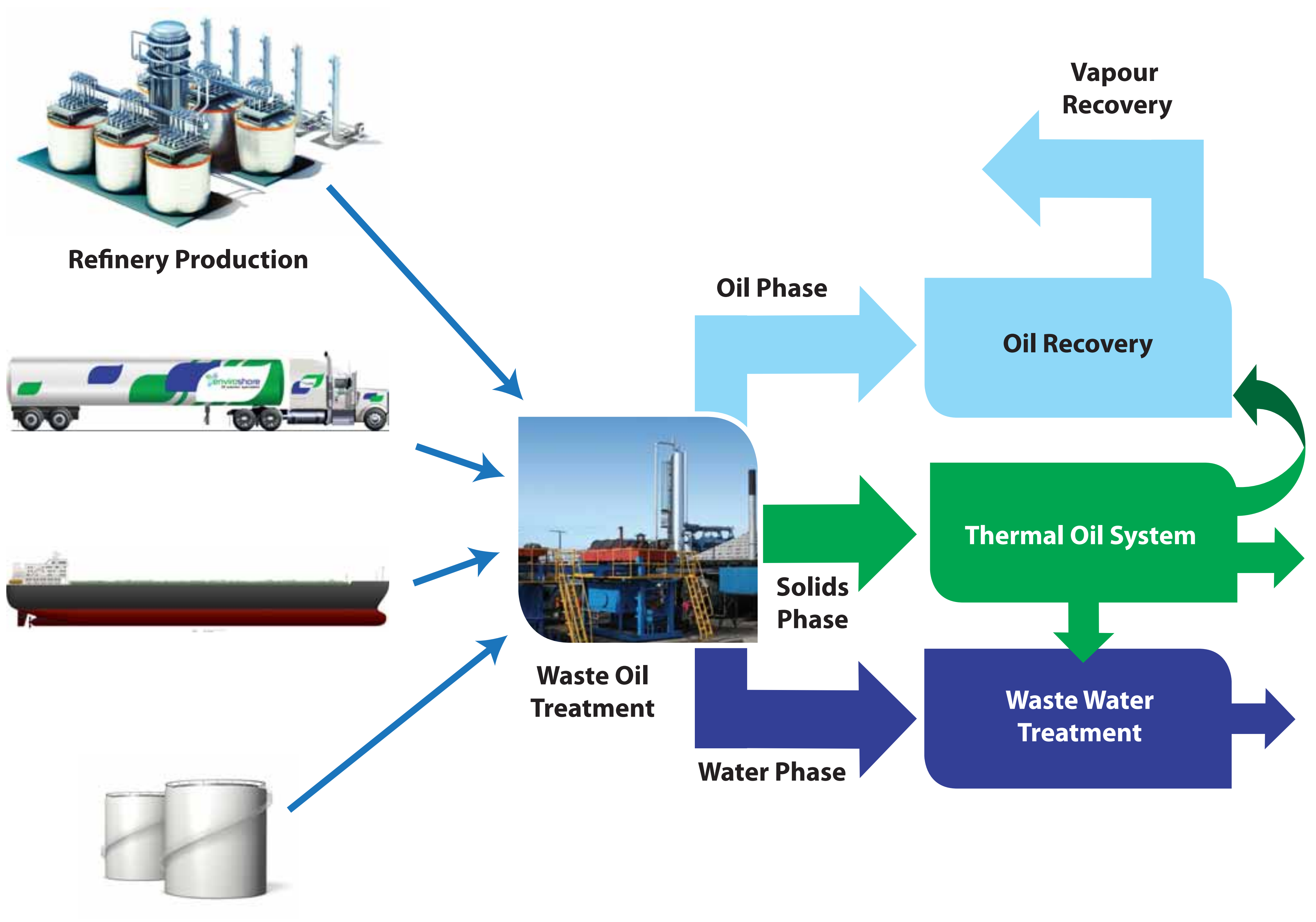
Goals and Objectives

- Phase One
  - Recover Product currently being stored in "Ponds".
  - Implement a recovery program.
  - Effectively bring the 'Pond" area to its natural state
  - Implement Quality control program
- Phase Two
  - Put up a bulk storage facility to handle and store the Recoverable Hydrocarbon material efficiently
  - Develop the downstream market, passing the benefit back to the refinery
- Phase Three
  - Utilizing the revenue generated from the sale of recovered hydrocarbon oils, invest in a BLABO system. This technology will thus improve the turn-around-time of the tank cleaning program for the refinery. Whilst working to our common vision of bring the latest oil recovery technologies to the refinery

Process Diagram in Brief



Marine Oil Pollution Control

Objective:
- Provide Solution: Establish an offshore oil pollution control company

Rationale:
- World Class: Deliver a world class product to provide effective pollution control

Sources of Major Spills
- Exploration and production well blow-outs (more likely in deep water)
- Offshore / sub-sea pipeline rupture
- Tanker and merchant marine vessel grounding
- Tanker collision (with other ships or platforms)
- Transfer spillage
- Release of contaminants from historically sunk ships

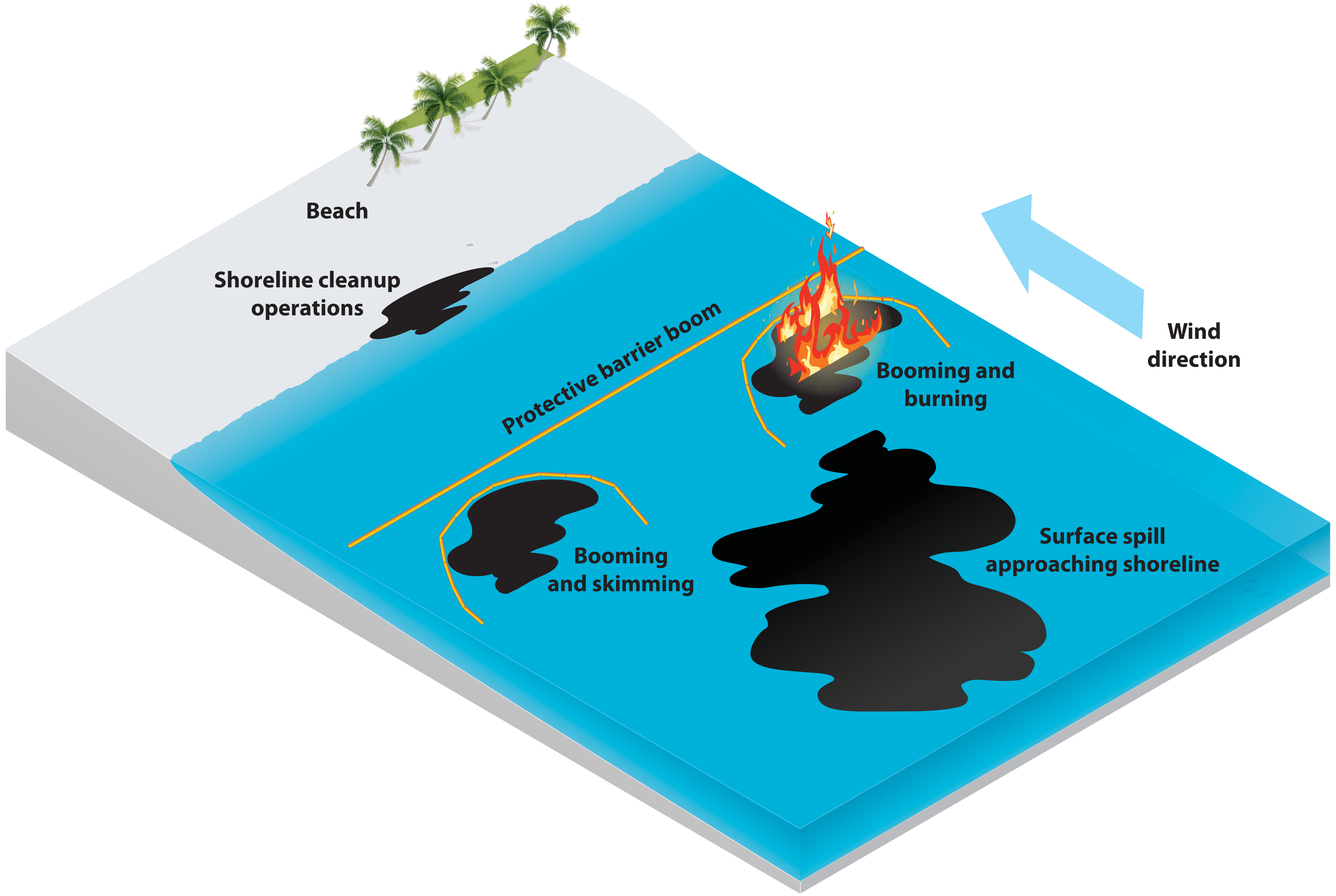
Objective
Providing an offshore pollution control capability that delivers:
- Effective, in-country, capacity to deal with offshore Tier 2 and Tier 3 oil spills
- Effective shoreline protection and clean-up
- Cross border capability
- World class standards
- A rapid response time (using local bases)
- Cost effective (cost sharing)
- A pollution solution for regional governments.
- Complimentary with global oil pollution control companies.

Philosophy for dealing with oil spills close to source and on the surface

- Rapid and efficient response
- Containment at source of spill
- If not possible, containment while spill is still waterborne
- Cleanup of spill while still waterborne
- If cleanup of water is not possible deflect oil to sacrificial beach/ area
- Operate within command and control structures as per the National Oil Spill Contingency policy

Near Shore Model

No dispersant used within 2000m of shoreline or in less than 20m of water depth. Use of chemicals in the near shore environment will only be considered in extreme cases and only after a "Net Environmental Benefit Analyses" (NEBA) has been completed.



What does this mean?

The first wave rapid response capacity equipment and personnel need to be held in locally (remote sensing - airborne and satellite, initial airborne and seaborne dispersant application, booming, burning and skimming, shoreline protection and cleanup)

The Second wave response can / will be flown in from regional / international bases (subsea dispersant application, subsea cap, relief well drilling, large scale airborne and seaborne dispersant application plus additional booming, burning and shoreline cleanup capacity)

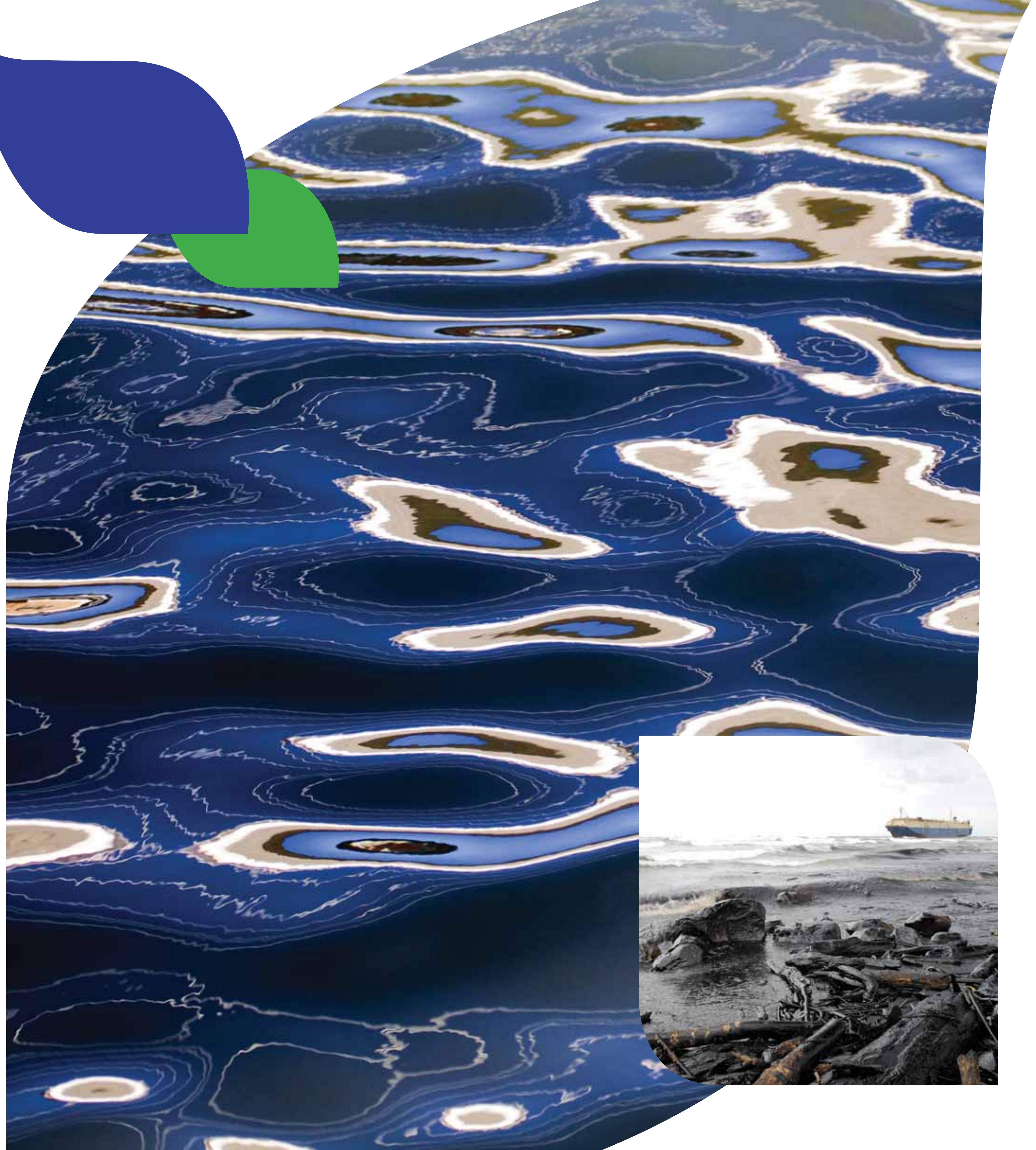
What will the OPC platform look like?

Equipment supply base warehouse (2 500m^2 - 3000m^2) with offices in your country

- Booms and skimmers
- Dispersant stocks
- Airborne and seaborne spray rigs (deployed on local vessels and aircraft)
- Oil storage and transport tanks
- Shore protection and cleanup equipment
- Remote sensing pods
- Producer and explorer PSV's available fo equipment deployment
- Core skilled personnel
- Training facilities

Capabilities

- At sea mechanical collection (booms and skimmers)
- Dispersant application (seaborne with potential airborne capacity)
- Burning
- Booming sensitive coastal environments
- Beach clean-up
- Airborne surveillance and monitoring, including remote sensing
- Capability
- Spill trajectory modeling



Boom Deployment

Containment at source of spill

If cleanup on water is not possible, deflect oil to sacrificial beach/area

Absorbents

BENEFITS

- 100% Organic, Non-toxic, biodegradable, self-generating in its natural state
- Better bio-remediation due to added nutrients in product
- Unique encapsulating qualities
- Eliminates up to 90% of VOC's (Volatile Organic Compounds) and LEL's (Lower Explosive Limit)
- Lightweight and easy to apply
- Powerful wicking action

The environmentally sound solution for oil and chemical spills, SpillSorb is the number one choice when it comes to treating chemical and hydrocarbon spills. SpillSorb has a 'barrel' cell structure that encapsulates pollutants not found in other products. SpillSorb is also non-toxic, biodegradable and self-generating in its natural state, with no danger to the environment.

The Client Value Proposition

- Real pollution control capacity
- Meaningful response times – Rapid response exponentially reduces the cost of spill remediation
- Cost effective – costs are shared across all clients
- No upfront capital expenditure
- True satisfaction of legislative, moral and triple bottom-line requirements
- Multiple bases deliver a true tiered response
- Retaining an operator experienced in executing in the African environment
- Lowering insurance related costs
- A standardised equipment base across East and West Africa
- Government backing
- Twins with multi-nationals oil companies and other pollution control initiatives (sub sea dispersant application, capping, capture, large airborne dispersant application capability and burning

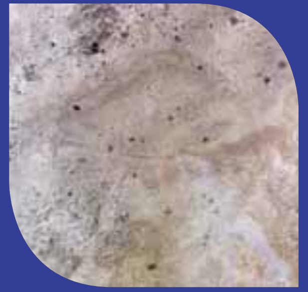
The Commercial Model

The business will be funded by either:
Preferred option: Government levying a pollution control tariff to oil and gas operators and explorers or....

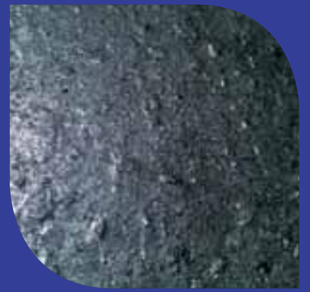
- Operators and Explorers will commit to supporting the platform
- Operators and Explorers will have access to the platform's emergency equipment, capacity and capabilities
- The platform will also be available to respond to emergencies outside of the oil and gas sector on a accounted cost basis
- ENVIROSHORE will commit to specific service levels - response time, skimming capacity, storage capacity, holding of specific pollution control equipment etc
- Use of the capacity in an emergency will be paid for by the operator or explorer on a cost plus basis "Freeloaders" will be charged a punitive rate for use of the capacity in an emergency (subsidising paid-up clients)

Our Products

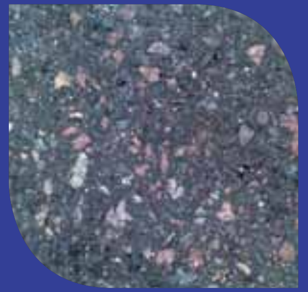
Before TAR-GONE



After TAR-GONE



Before OIL-GONE



After OIL-GONE

Before MICROBE SOIL TREAT

After MICROBE SOIL TREAT

Before ECO ACTIVE

After ECO ACTIVE

Tar-Gone

A Superb Bitumen Cleaner/ Degreaser. TAR-GONE is a revolutionary new bitumen water-soluble cleaner and degreaser, which in its form is non-hazardous, user friendly and will not harm the environment.

Microbe Oil Gone

Microbe Oil Gone is a dual-purpose oil/ fuel degreaser and bio-degrader that readily breaks down all hydrocarbons. The natural non-pathogenic bacteria, enzymes (micro-organisms) present in Microbe Oil Gone naturally biodegrades the contaminant to harmless bi-products of H2) and carbon dioxide on site, in a short time frame.

Oil-Gone (Water Based)

A powerful, water based cleaner for hydrocarbons and greases based on natural citrus oils. Oil-Gone WB is an environmentally friendly product, used for the rapid removal of heavy oils and greases on a wide range of floors and surfaces without damaging paintwork.

Microbe Soil Treat

Microbe Soil Treat is a dual-purpose oil/fuel absorbent and bio-degrader that readily absorbs all hydrocarbons with at least 1.7 times its own weight. The natural non-pathogenic bacteria, enzymes (micro-organisms) present in Microbe Soil Treat naturally degrades the contaminant to harmless bi-products of H_2O and carbon dioxide on site, in a short time frame.

Eco Active (Low Foam & High Foam)

A powerful water based cleaner for hydrocarbons and heavy greases using the latest technologies for a optimized degreasing effect.

Algae-Gone

A stabilized-glutaralderhyde sanitizing solution for the elimination of algae, bacteria, viruses, fungi, moulds and spores.

ALGAE-GONE is a PH-Neutral clear liquid that offers the most effective chemical for cleaning algae, bacteria, viruses, fungi, moulds and spores. In addition, its 'user-friendly' nature reduces the operational costs and eliminates the after-effects of corrosion.

ALGAE-GONE works instantly on most surfaces. After spraying an area, the germs are immediately removed simply by pouring water over the affected area. The results are immediately apparent.

The **Earthwize** range of spill cleaning products is distributed by **Enviroshore**

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- Better bio-remediation due to added nutrients in product
- Unique encapsulating qualities
- Eliminates up to 90% of VOC's (Volatile Organic Compounds) and LEL's (Lower Explosive Limit)
- Lightweight and easy to apply
- Powerful wicking action

ECOCARB HYDOCARBON ABSORBENT

A quality, locally made hydrocarbon absorbent for the volume user.

BENEFITS:
- 100% organic
- Non-toxic and biodegradable
- EcoCarb is manufactured by treating the raw organic material with a patented process
- Increased porosity aids in absorption resulting in a quality product that does not leach
- EcoCarb is capable of absorbing 0.65 litre of a light crude oil per litre of product

ACIDCARB HYDOCARBON ABSORBENT

The easiest, most general purpose hazardous liquid absorbent.

BENEFITS:
- Non flammable, fully oxidised, stable, inert and 100% naturally occurring mineral
- Non-carcinogenic and non-toxic - harmless to plants, animal life and the environment
- Absorbent - Adsorbant
- VOC-LEL reduction
- Easy to use – cover the spill. Absorbs the spill on contact
- No conditions to avoid, incompatible materials or hazardous decomposition products
- AcidCarb is not biodegradable; the products it absorbs may be bio-degradable

BIOCARB BIO-REMEDIATION VIA FUNGAL CULTURE

Provides end users with an alternative solution to land-fill when dealing with petrochemical and chlorinated hydrocarbon contaminated solids.

BENEFITS:
- 100% natural-absorbent
- Bio-remediation has been internationally researched for over 30 years and is supported via the EPA-USA and DWAF in SA under consultation
- One bag treats 1 cubic metre of contaminated solids
- Manufactured using a patented process utilising fibrous recycled plant material from the vegetable oil industry
- Contains a diverse range of microbes, bacteria and filamentous fungal species that effectively degrade metabolic bi-products during the remediation

FUNGICARB LIQUID BIO-REMEDIATION

For the treatment of Petrochemical and Chlorinated Hydrocarbon contaminated water and soil – inaccesible places.

BENEFITS:
- Contains three fungal cultures and surfactant producing bacterium
- Environmentally friendly
- Introduces nutrients and hydrocarbon - degrading microbes to contaminated water/inaccessible solids
- Promotes environmental conditions conducive to breaking down hydrocarbon based contaminants
- Cost-effective solution to environmental pollution

DECARB SOLVENT FREE MICROBIAL DEGREASER

An extremely efficient general-purpose water-based degreasing solution.

BENEFITS:
- High performance liquid remediation solution that works instantly
- Provides a unique blend of natural surface active agents that promote wetting, solubilisation and emulsification of various organic chemicals
- Environmentally friendly, non-flammable, easily bio-degradable and non-corrosive
- Highly concentrated, reducing clean-up costs
- For use on hard surfaces – safe to use on asphalt, as an engine cleaner and general cleaning solution

SPILL KITS

- Oil Only: (Hydrophobic) All hydrocarbons
- Universal: (Hydrophilic) Water, Oils, Solvents and Coolants
- Anti Static
- Hazardous: Chemical, Acid or Base Compounds
- Bio-Remediation: (Microbes or Fungal)

SPILL PLUG

BENEFITS:
- Sticks to dirty, crumpled surfaces even with fluid flowing out. Confines the spill – minimises clean-up
- Recommended for mostly hydrocarbons and many water-based materials
- Non-toxic and non-flammable
- High-absorption polymers in a blended bentonite base forms an immediate seal
- Has been used for over 15 years by fire departments, transportation companies and industrial operations


The **journey** to the **future**
begins today!


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